



YATING Arts

當AI遇見數位音樂： 創作工具的 介紹與應用

講者 葉胤呈 蕭文逸



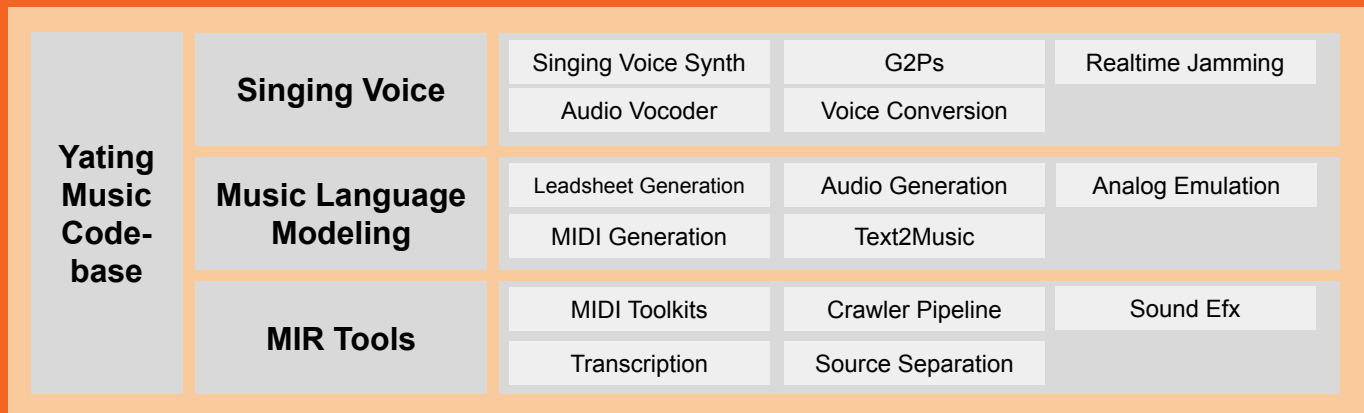
葉胤呈

**Team Lead,
Taiwan AI Labs**



蕭文逸

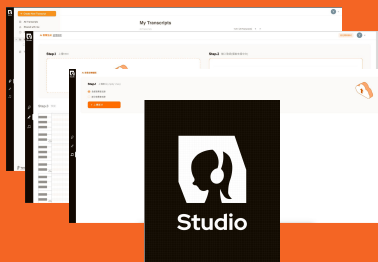
**Research Engineer,
Taiwan AI Labs**



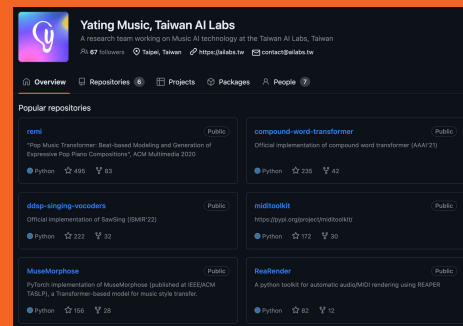
Projects



Products



Research







High Quality



Easy to Use



Affordable

What AI Can Do in Music Production



大綱



Artificial Intelligence 101

Virtual Instruments - Vocal

Transcription and MIDI Generation

Text2Sound

Audio Efx Modeling

Spatial Audio



Symbolic Domain



Audio Domain

大綱



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Symbolic Domain

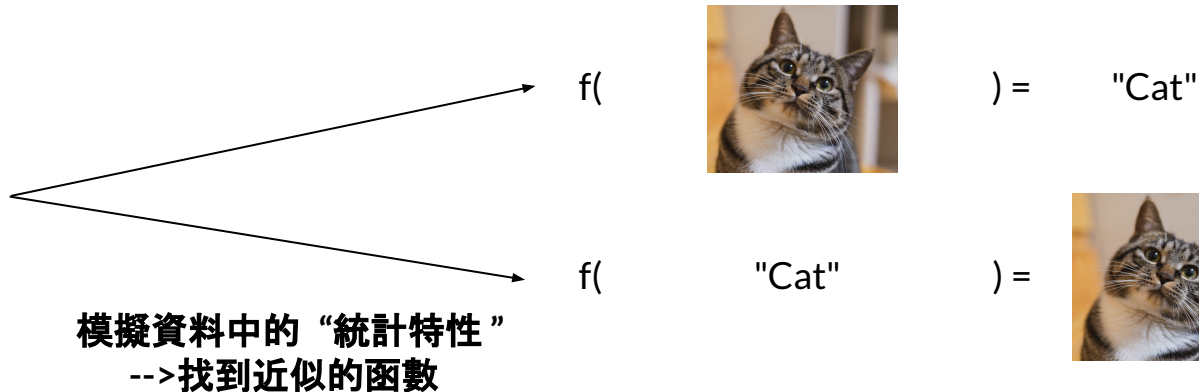


Audio Domain

Artificial Intelligence 101 - Concepts



資料
(圖像、序列 ...)



$f(\text{"Compose a piece in a style of J. S. Bach Choir"}) =$



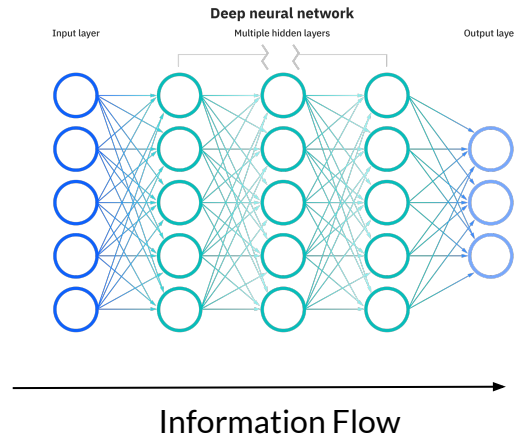
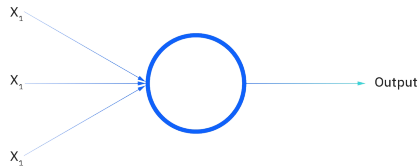
Note: Deep Bach
<https://arxiv.org/pdf/1612.01010.pdf>

作曲手法和詮釋想法也能看作是某些“統計特性”所構成

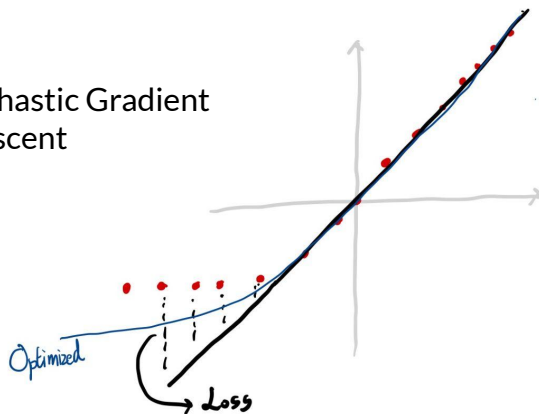
的
那要怎麼讓 AI “練習” 才能塑造這些統計特性呢？

Artificial Intelligence 101 - Terms (1)

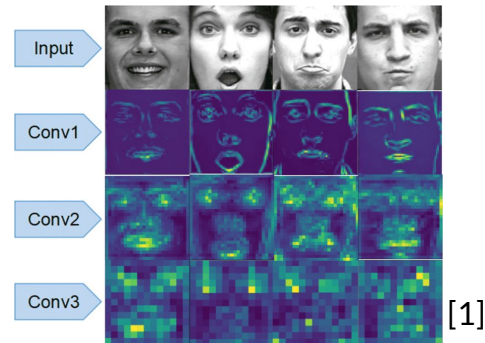
類神經網路 (Neural Network)



Loss and Stochastic Gradient Descent

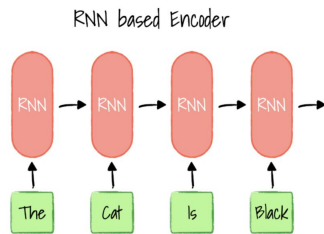


Feature Maps



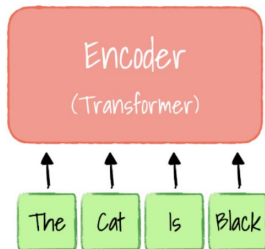
Artificial Intelligence 101 - Terms (2)

循環神經網路 (Recurrent Neural Network)

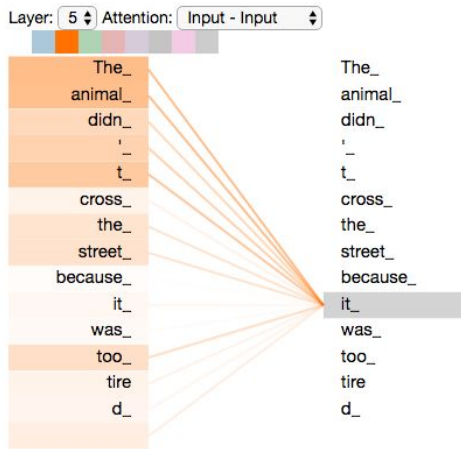
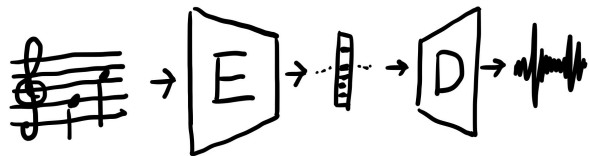


轉換器 (Transformer): 大型語言模型 (Language Model) 的基礎架構之一。

Transformer's Encoder



編碼器 (Encoder)、解碼器 (Decoder) 和 聲碼器 (Vocoder)

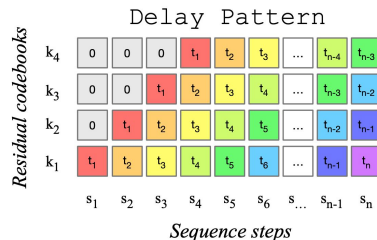


Artificial Intelligence 101 - What's the limit of AI in Music

大資料 + 大模型 (ChatGPT, MusicLM, MusicGen)



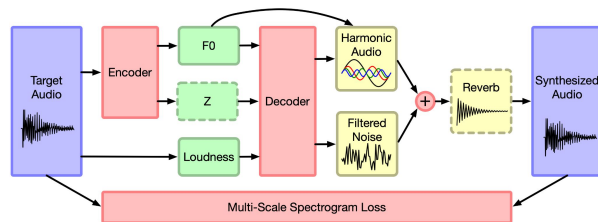
大資料要夠多模型才能夠頓悟 (Emergence)



MusicGen

<https://audiocraft.metademolab.com/musicgen.html>

小資料 + 設計精良的小模型 (DDSP, Sound Efx)



DDSP

<https://magenta.tensorflow.org/ddsp>

在音樂上,

資料要怎麼定義呢?

模型要怎麼設計才會好用呢?

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Artificial Intelligence 101

Virtual Instruments - Vocal

Transcription and MIDI Generation

Text2Sound

Audio Efx Modeling

Spatial Audio



Symbolic Domain



Audio Domain

Virtual Instruments - Vocal

Why Vocal

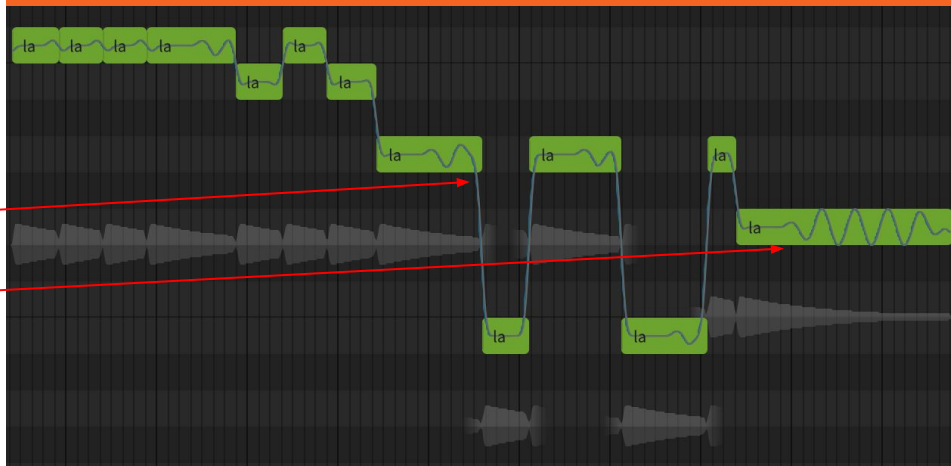
Why Vocal Hard:

1. Numerous Phoneme Combination
2. Pitch Transition
3. Various Expression and Techniques

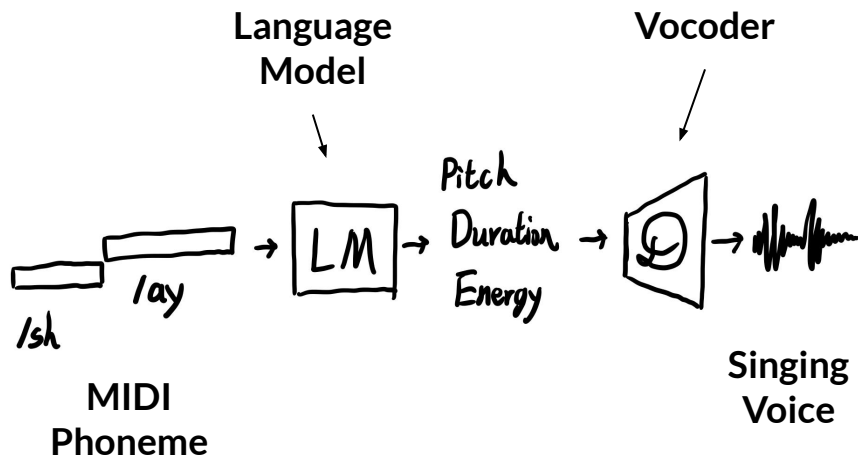
Singing Voice Synthesis

Voice Conversion

Real-Time Jamming



MIDI Singing Voice Synthesis



Vocaloid <-- Not Real, Hard to Use

Synthesizer V <-- Hard to Use

ACE Studio

(<https://ace-studio.timedomain.cn/>)

Yating AI Vocal (Ours)

陳珊妮
教我如何做你的愛人



真人演唱 vs AI Vocal



新進男演員

23歲

目前正與AILabs合作，
訓練他的AI人聲模型。



新人女歌手

19歲

預計於今年7月發行
第一支個人單曲。

MIDI Singing Voice Synthesis

Yating AI Vocal Demo

- Web Interface:

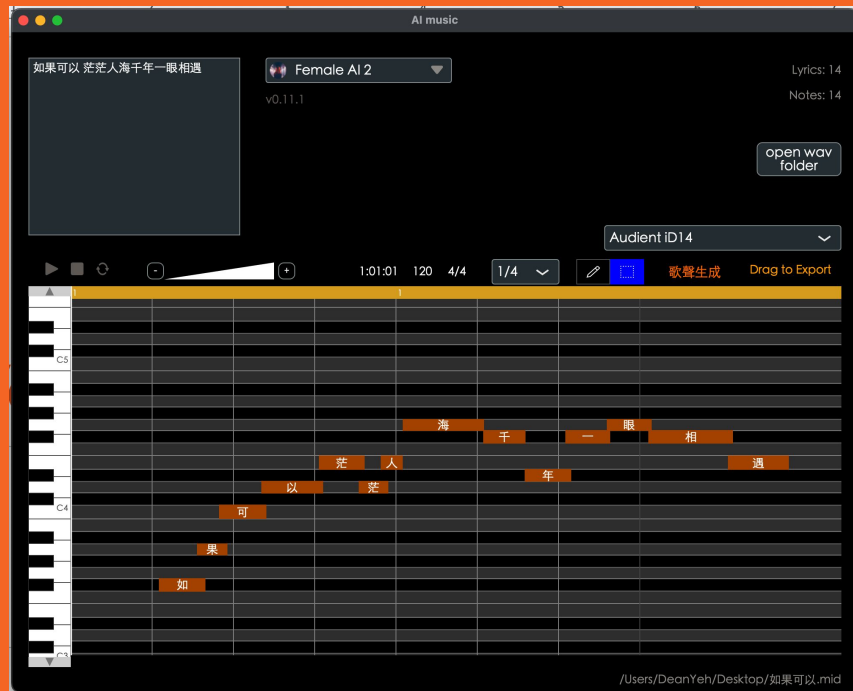
<https://music.yating.tw/music/voxai>

- App Interface:

<https://music.yating.tw/music/downloads>

討論:

1. 會希望Vocal還有那些功能呢？
2. 除了Vocal之外，還希望可以有什麼樂器呢？



Logic Project Demo: 如果可以 Covered by AI Vocal

Voice Conversion

目前AI可以做得多好

Supertone VC Demo: <https://supertone.ai/>

Resources

[App] Voicemod

<https://www.voicemod.net/>

[Github] so-vits-svc

<https://github.com/svc-develop-team/so-vits-svc>

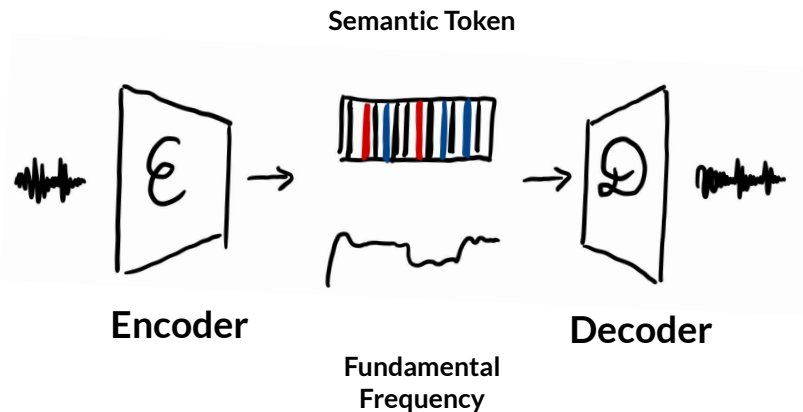
[App & Github] RVC-Project:

<https://github.com/RVC-Project/Retrieval-based-Voice-Conversion-WebUI> (AI孫燕姿和統神從這個來)

Voice Conversion

為什麼AI可以做得比較好呢

1. Naturalness
2. High Quality Scale Transportation



Voice Conversion in Action - Lullaby of Birdland

Input Audio:

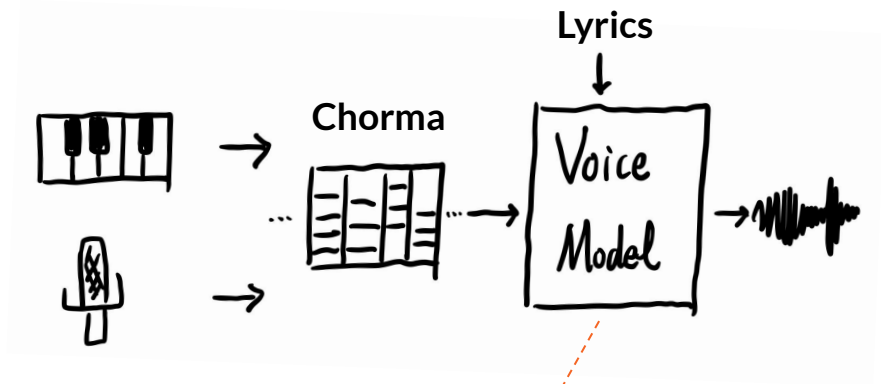


Target: Sandee

Voice Conversion Audio:



Live-Jamming



Conclusion and Discussion

長音的詮釋仍然有些缺陷

高頻噪音

有沒有用途呢？

控制項的設計

還會想要什麼虛擬樂器呢？

即時互動式重要的嗎？

大綱



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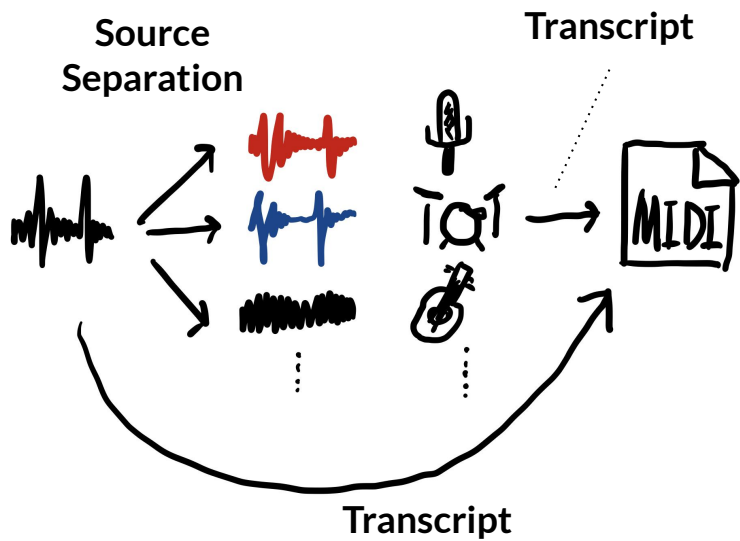


Symbolic Domain

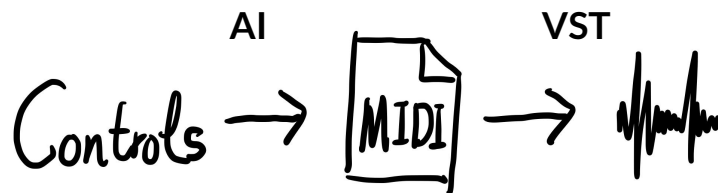


Audio Domain

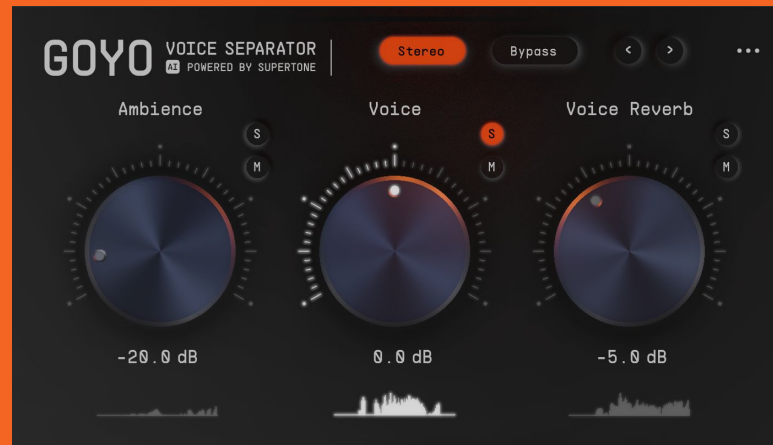
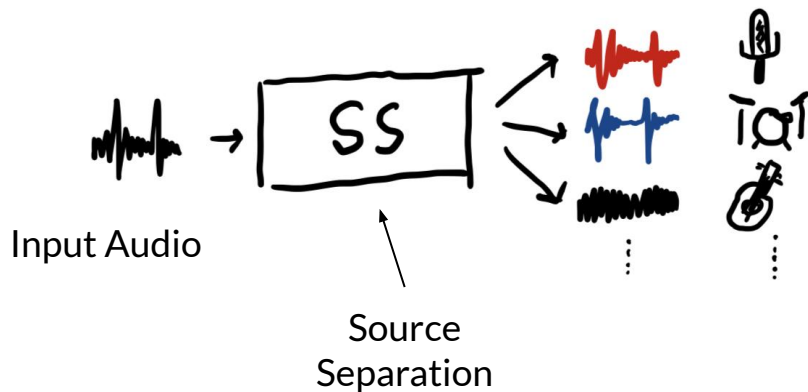
Transcription



Generation



Source Separation



Resources:

[Voice Separation VST] <https://goyo.app/>

[Moises App] <https://moises.ai/>

[Web] <https://huggingface.co/spaces/akhaliq/demucs>

[Github] <https://github.com/facebookresearch/demucs>

Transcription

[Github] Piano Transcription^{Note 1}

https://github.com/qiuqiangkong/piano_transcription_inference#piano-transcription-inference

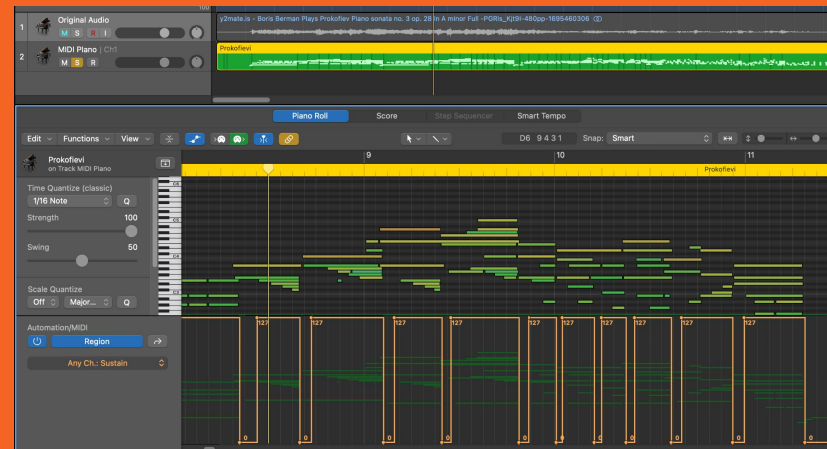
[VST] NeuralNote

<https://github.com/DamRsn/NeuralNote>

[Web] Basic Pitch Web Demo

<https://basicpitch.spotify.com/>

(Optional) Resynthesized Example



NOTE 1: Piano-e-competition and Maestro Dataset: <https://www.piano-e-competition.com/>

MIDI Generation- Piano

Blog and Colab Link:

<https://magenta.tensorflow.org/music-transformer>

Listen to Music Transformer:

<https://magenta.github.io/listen-to-transformer>

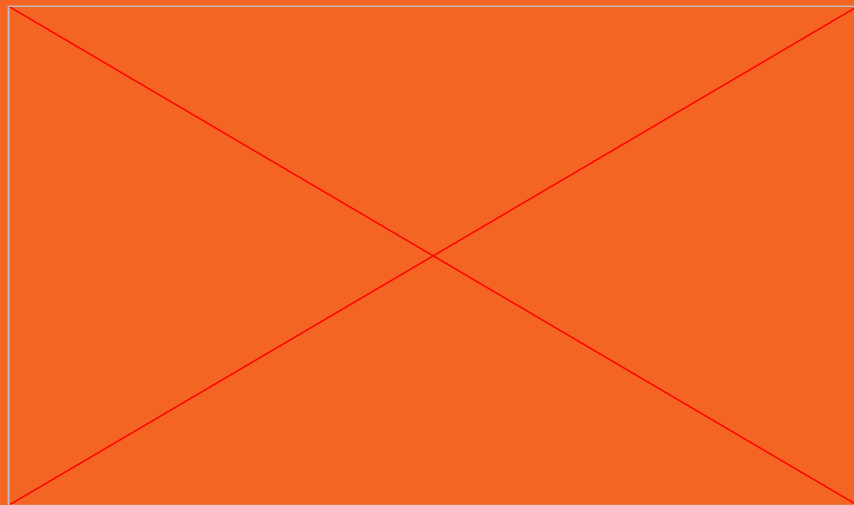
Chord Condition: CM | GM | DM | Em |



Original Piece in Jazz Style :



Debussy's "Clair de lune" Continuation:



— ▲ How AI "think" when plays piano

MIDI Generation- Drums

[Dataset] Groove MID Dataset

<https://magenta.tensorflow.org/datasets/groove>

[VST] Magenta Studio

<https://magenta.tensorflow.org/studio/>



Conclusion and Discussion

生成沒有結構

只有敲擊樂器做的比較好, 管弦樂的 Automation 都拉得不是很好

會想要用在什麼地方呢？

還會想要什麼樂器呢？

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Artificial Intelligence 101

Virtual Instruments - Vocal

Transcription and MIDI Generation

Text-to-Music

Music-Knowledge-to-Music

Text-to-SFX



Symbolic Domain



Audio Domain

Text2Music

Text-to-Music

使用文字(text, or prompt)描述來創作音樂

A punchy double-bass and a distorted guitar riff



Lofi slow bpm electro chill with organic samples



Smooth jazz, with a saxophone solo, piano chords,
and snare full drums



A grand orchestral arrangement with thunderous
percussion, epic brass fanfares, and soaring strings,
creating a cinematic atmosphere fit for a heroic battle



Text-to-Music

使用文字(text, or prompt)描述來創作音樂

Prompt Engineer (提示詞工程) in Text2Music

樂器

曲風

形容詞

A grand orchestral arrangement with thunderous percussion, epic brass fanfares, and soaring strings, creating a cinematic atmosphere fit for a heroic battle

Smooth jazz, with a saxophone solo, piano chords, and snare full drums

DEMO: <https://f89fbe1d9217364aa8.gradio.live/>

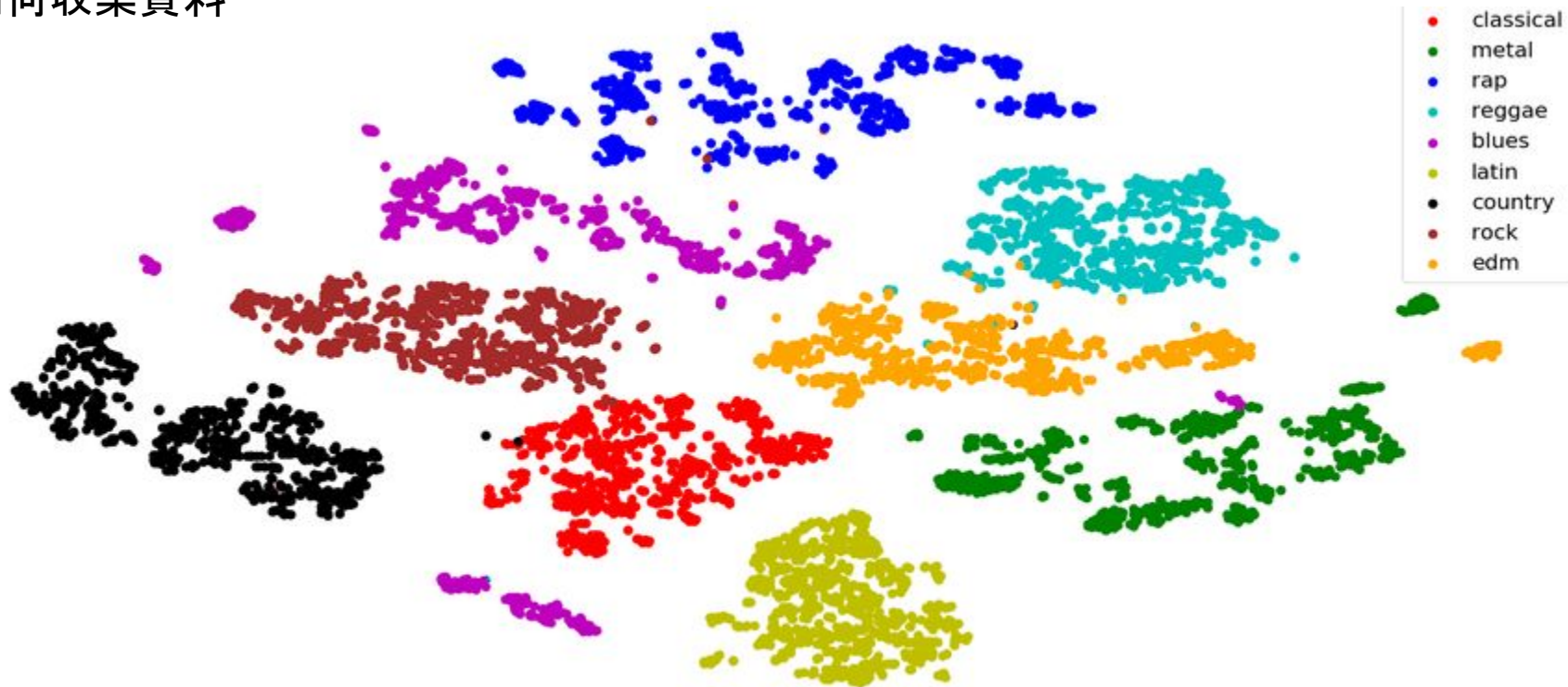
Text-to-Music原理

- 技術核心: Large Language Model (LLM)
- 2萬小時訓練資料, 囊括各種曲風
- 每一筆資料以”文本輸入”來訓練模型產生”音樂”
- 資料來源
 - Shutterstock
 - Pond5



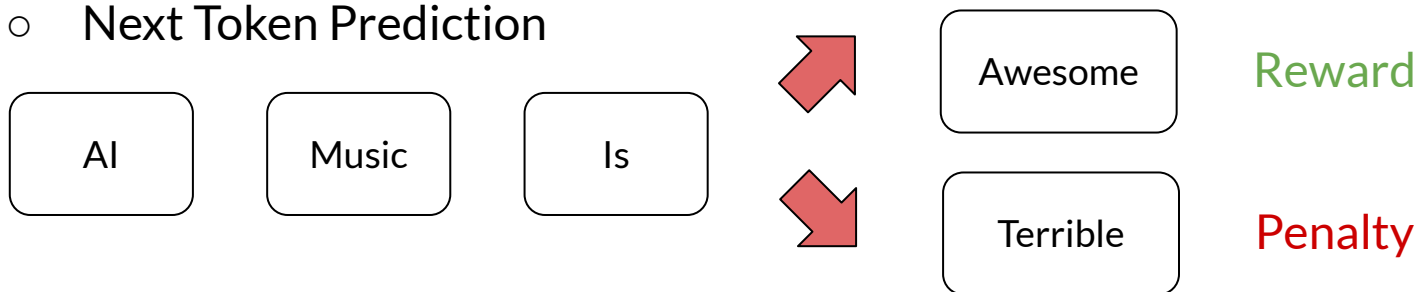
Text-to-Music原理

如何收集資料

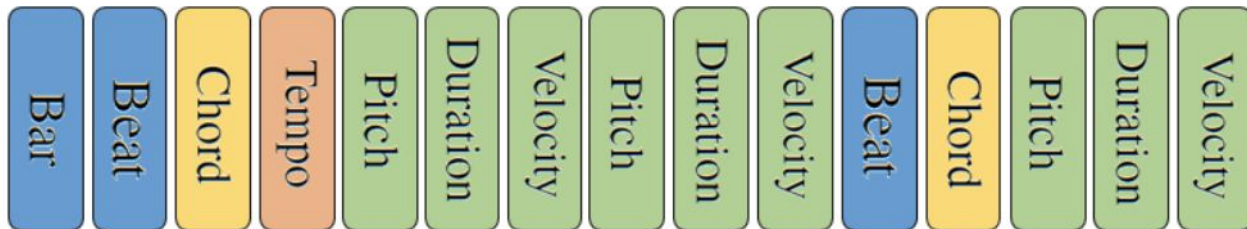


Text-to-Music原理: LLM

- LLM原理, Chapgpt
 - Transformer
 - Next Token Prediction



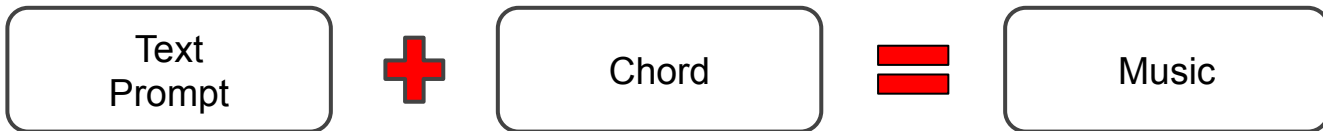
- Model如何看/認知音樂



Music-Knowledge-to-Music

- 文本產生音樂的問題
 - 語言選擇 (中文, 英文, ...)
 - 無法精確控制
 - 受眾為普羅大眾
- 利用樂理來產生音樂
 - 西方音樂理論
 - 曲風, 編制, 速度, 調性, 韻律, 和弦, 旋律
 - 受眾為音樂愛好者與從業者
- 不同的輸入端
 - 文本 -> 樂理 -> 音樂
 - 影片 -> 樂理 -> 音樂

Example I: 和弦進行



A grand orchestral arrangement with thunderous percussion, epic brass fanfares, and soaring strings, creating a cinematic atmosphere fit for a heroic battle

Am G F E



DEMO: <https://4d333cc40928e0af46.gradio.live/>

Example II: 旋律



An energetic hip-hop music piece, with synth sounds and strong bass. There is a rhythmic hi-hat patten in the drums.

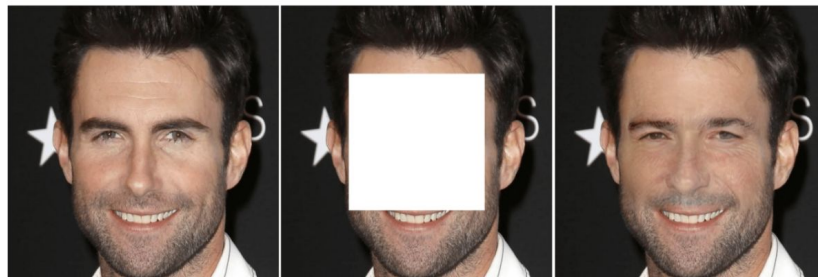
Bach



DEMO: <https://audiocraft.metademolab.com/musicgen.html>

應用延伸

1. Inpainting:



Masked

Filled

Intense, drama, climax, avengers, dark knight, man of steel

Motivational, driving, anthemic, uplifting, positive, underscore



應用延伸

2. Continuation:

Sad, emotional, dreamy, reflective,
ambient, pensive

Upbeat, optimistic, joyous, cheerful,
light, happy

Masked

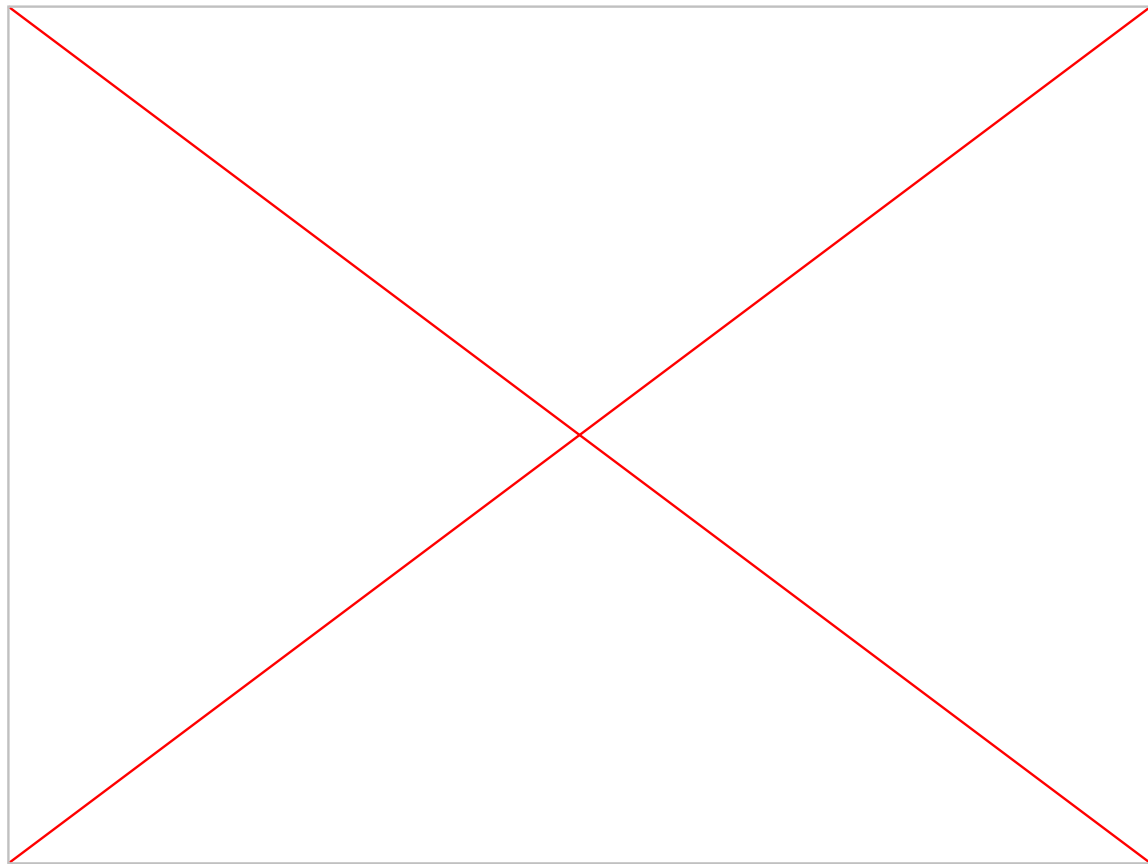


Filled



應用延伸

3. Editing



Music Understanding System

- Prompt: Describe the contents of the provided audio in detail.



- AI:

“As I listen to the audio clip, I hear a rock song with a lively and energetic feel. The song starts with a guitar playing a short riff, followed by a drum beat. The guitar continues to play a catchy melody, while the drums maintain a steady rhythm throughout the clip. The tempo of the song is moderately fast, with a beat that makes you want to move and dance...”

- Demo: <https://storage.googleapis.com/music2text-public/index.html>

Text-2-Sound-Effect (SFX)

使用文字(text, or prompt)描述來創作音效

A forest of wind chimes singing a soothing melody in the breeze.



Soft whispers of a bedtime story being told.



The splashing of water in a pond.



DEMO: <https://audioldm.github.io/audioldm2/> (2023)

Text-to-Music的潛力

- 跳過許多步驟(MIDI Mockup, 錄音, 混音, 編曲)直接生成成品
 - 普通人也能做出接近專業水準的作品
 - 專業人士可以快速發想點子
- 利用形容詞來創作
- 生成世界上不存在的音樂
 - 避免版權問題
- 取代素材網

- Ex:

Jazz

Saxophone
Solo

A min

Drum Brush

.....

Text-to-Music的瓶頸

- 版權
 - 爬蟲
 - 使用licenced data來訓練是否為侵權?
- 算力資源
 - 訓練: 32 GPU x 5 days
 - 使用: 1 GPU標配
- 音質
- 更精確控制項
 - 轉位, dry/wet mixing, ...
- 不確定性
- 結構

我們失敗的案例

Sad Lofi



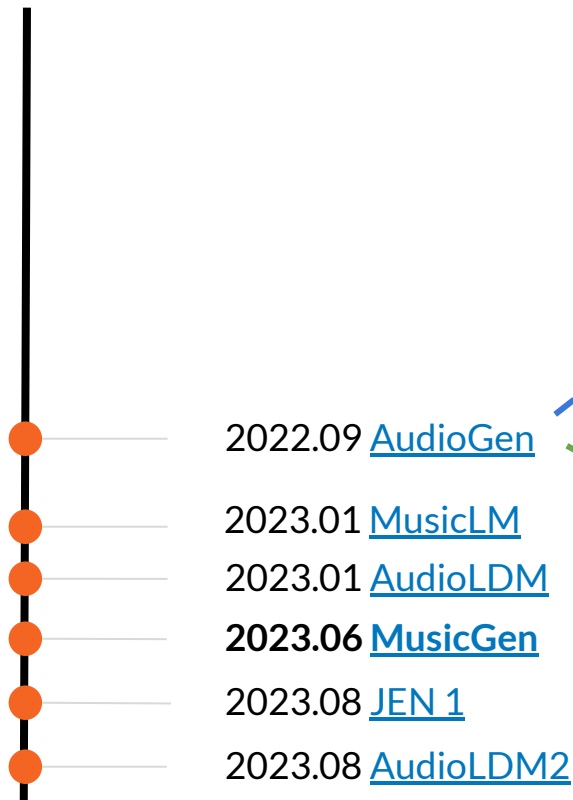
Chill Jazz



Long Sample



AI音樂時間軸



AI繪畫時間軸



Quality

Date

Text-to-Music的潛力

- 教學
- 結合AI Vocal & AI Lead sheet生成
 - AI詞曲創作體驗

By YCYeh

The image shows a web-based interface for AI music generation, divided into two stages. Stage 1, '詞曲創作' (Lyrics and Music Creation), includes a text input for lyrics (currently containing cat emojis), a key selector set to 'E', a tempo slider set to 104.2, and a chord progression selector set to 'ChordProgression S'. Stage 2, '描述編曲' (Describe Arrangement), is currently disabled. A 'Preview' section shows a play button and a 0:15 duration. A 'Song ID' section displays a unique identifier. A large '生成' (Generate) button is at the bottom.

Stage 1: 詞曲創作 (做到喜歡的詞曲之後就可以拿著Song ID進stage2編曲了)

Stage 2: 描述編曲 用英文(會等很久, 大概一到兩分鐘)

輸入歌詞 創造詞曲 (用空白或換行當作歌詞的段落 只能打中文 建議四到八句歌詞)

貓貓貓貓貓貓貓貓貓貓貓貓

Key
Select the Key of the Leadsheet
E

Tempo
Choose the tempo of the leadsheet
104.2

Chord Progression
Select one chord progression
ChordProgression S

詞曲Preview
0:15

Song ID
45fc6852-849d-11ee-a135-4abdc8b7f928

生成

DEMO: <https://5165b79160750146ad.gradio.live/>

What AI Can Do in Music Production

作

Virtual Instruments (MIDI/Audio)

編

Text2Music

混

Audio Efx / Spatial Audio

解放雙手和時間
讓靈魂往的音樂去吧！

Slide Link:

<https://www.slideshare.net/wayne391/20231117-ai-music-intropdf>



Contact:

- yyeh@ailabs.tw
- wayne391@ailabs.tw

Backup

大綱



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Audio Domain

大綱



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Audio Domain

AI Audio Effect

- AFX -

$$y = f(x, c_t, c_g)$$

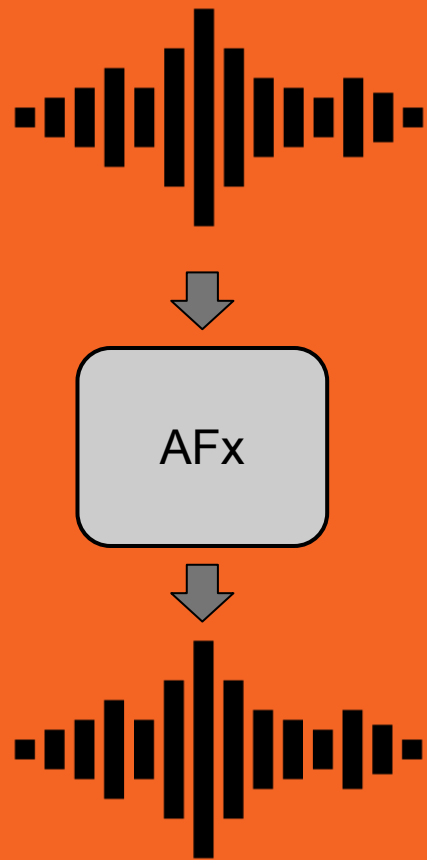
x: Input signal, M channel

y: Output signal, N channel

c_g : Global condition

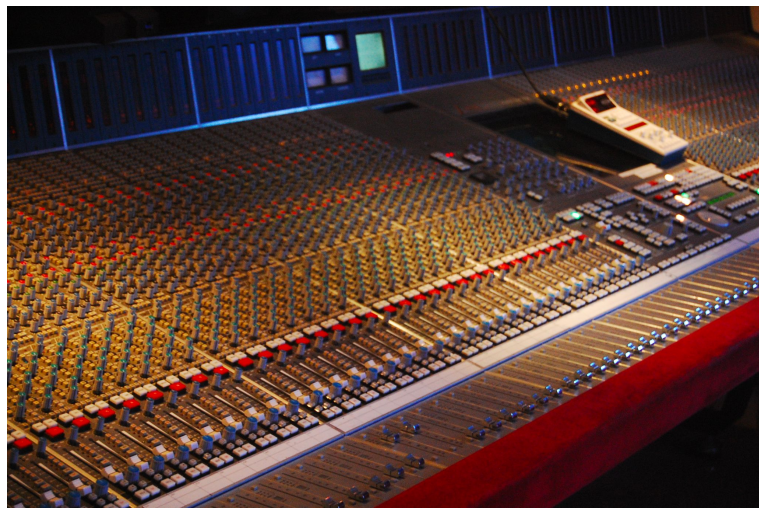
c_t : Local condition

重點是函數本身！



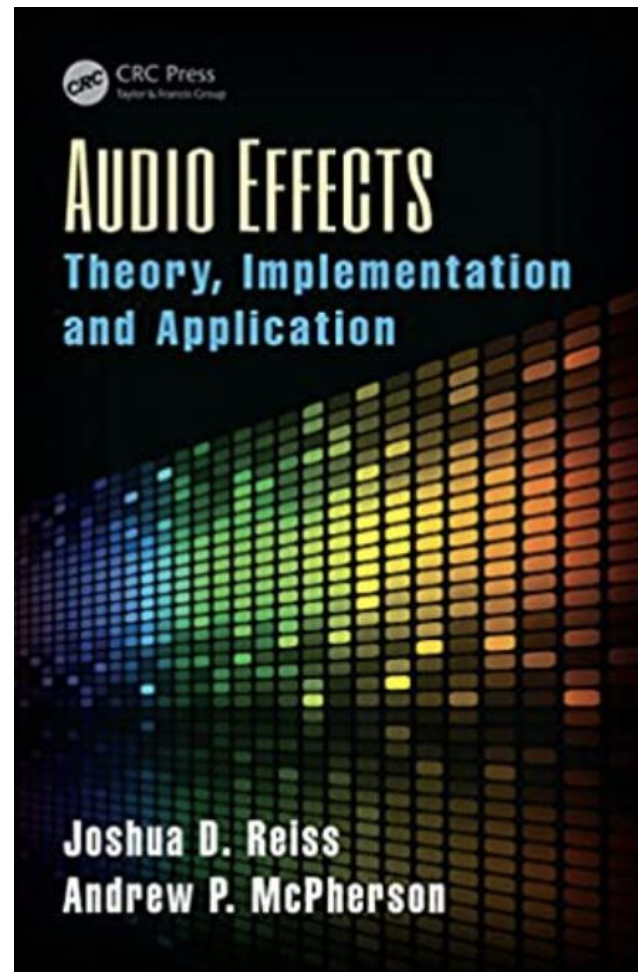
Audio Effect

- 混錄音 | 母帶 | 吉他
- AI 可以做什麼？
 - 把硬體數位化
 - 降低成本/易於維護
 - 保留參數 (as presets)
 - 推廣好的聲音與器材
 - 保留類比感
 - 學習操縱參數
 - 設計Routing
 - 降低學習曲線



Audio Effect

- TextBook for Engineering & Coding
 - **Audio Effects** by Joshua D. Reiss
 - Github: [juandagilc/Audio-Effects](https://github.com/juandagilc/Audio-Effects)
- Scenarios
 - EQ - Parametric EQ, Graphic EQ
 - Dynamics - Compressor, Limiter, Expander
 - Distortion - Overdrive pedal, Amp, Saturator
 - Reverb - Chamber, Hall, Room
 - Delay - Spring delay, Tape delay
 - Modulation - Flanger, Chorus, Phaser
 - Spatial - Stereo imager, Mid/Side processor
 - Others - Noise reduction, Pitch-correction



Audio Effect

- AI 比起傳統方法好在哪邊？
 - 只要準備輸入/輸出資料讓機器學就好
 - 傳統需分析複雜電路
 - One-by-one
 - 不同種類的AFx可以用一個模型解決
 - EQ / Compressor / Delay...

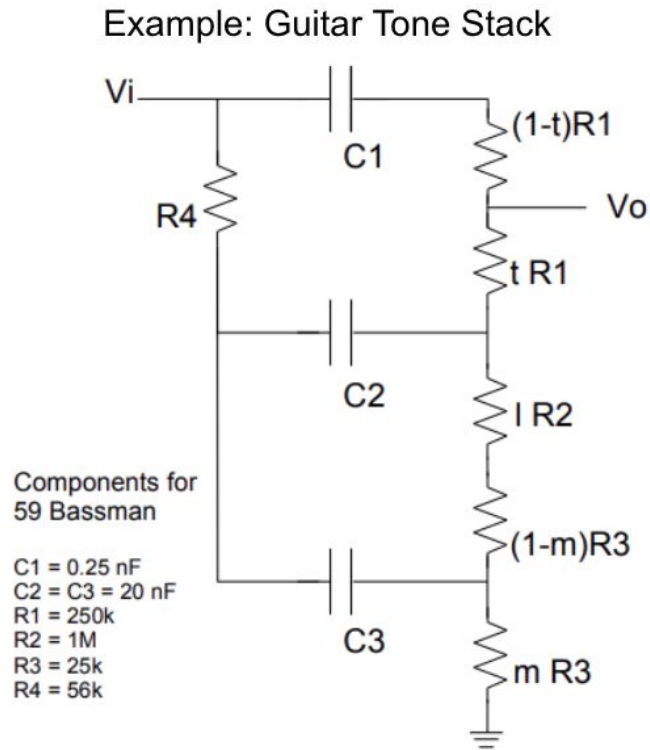


Figure 1: Tone stack circuit with component values.



QUAD CORTEX

PLUGINS

CAREERS

CORTEX CLOUD



Archetype: Tom Morello

The iconic sounds of a modern guitar legend.

[Learn more](#)

AI目前可以模擬得多好？

- A/B Testing
- Example: Soldano SOL-100



開源專案

- 大家都可以動手訓練自己的音色
- 基本C++ JUCE框架的理解

<https://github.com/GuitarML/SmartGuitarAmp>



Keith Bloemer

GuitarML

Unfollow

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Anyone can get great guitar tone using modern computing technology. GuitarML is a community for advancing machine learning for use in electric guitar plugins.

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NeuralPi: 動手做硬體

Requirements

- Raspberry Pi 4b
- HiFiBerry DAC + ADC
- Elk Audio OS
- NeuralPi VST3 plugin



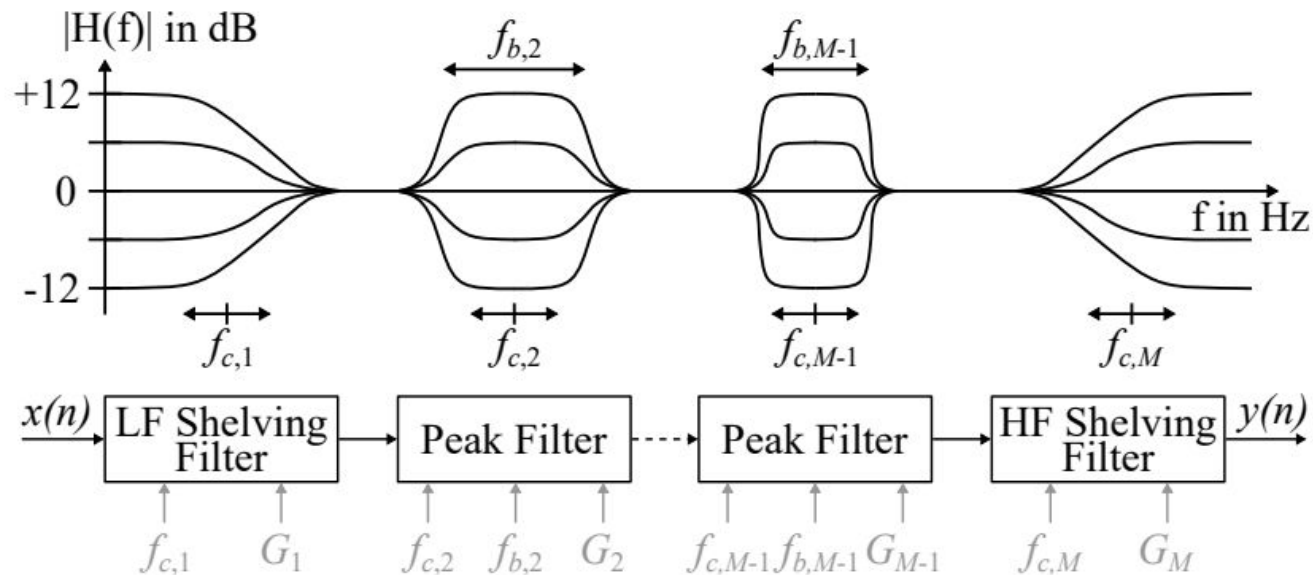
<https://github.com/GuitarML/NeuralPi>

Limitations

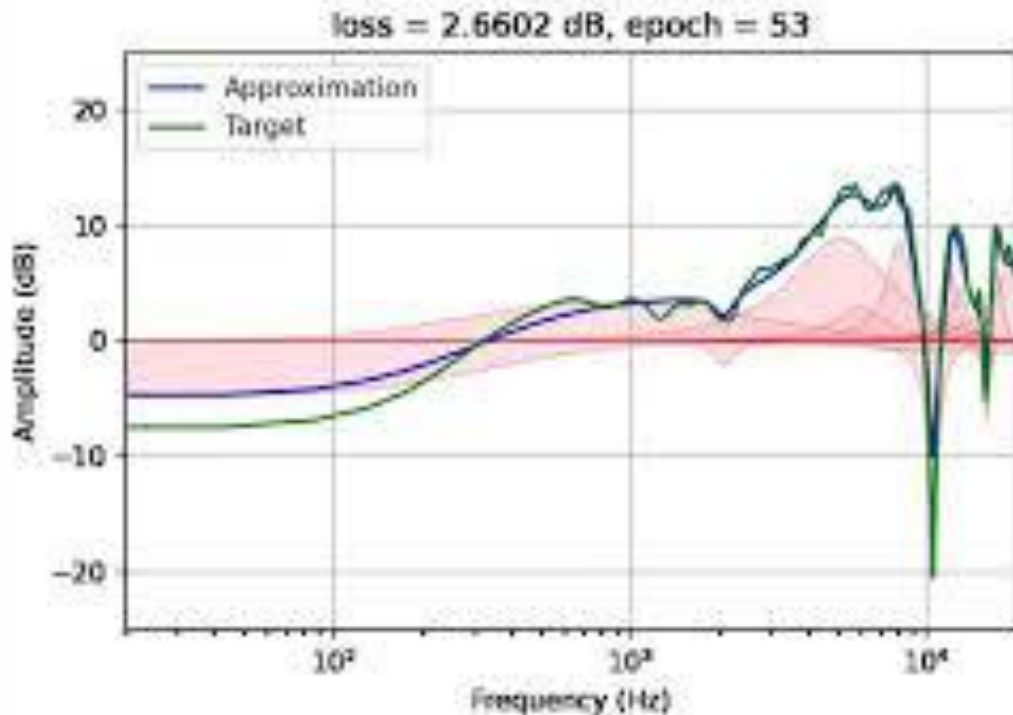
- 做動過長的效果仍然不夠好
 - Delay, Reverb, ...
- 運算資源消耗龐大
 - 黑箱 => 參數多 運算複雜
 - 48KHz, 96kHz ...
- 動態仍然有限
 - 塑膠感
- 傳統方法 - High Gain 優 | Low Gain 劣
 - Metal: Dream theater
- 深度學習 - High Gain 劣 | Low Gain 優
 - Funk: Cory Wong



AI Learn to Control Knobs



讓AI學習使用Parametric EQ去擬合Reference Track的Spectrum



SRC: yoyololicon, PhD student at @c4dm, Queen Mary University of London

好物分享

- Jatinchowdhury18
@Stanford CCRMA

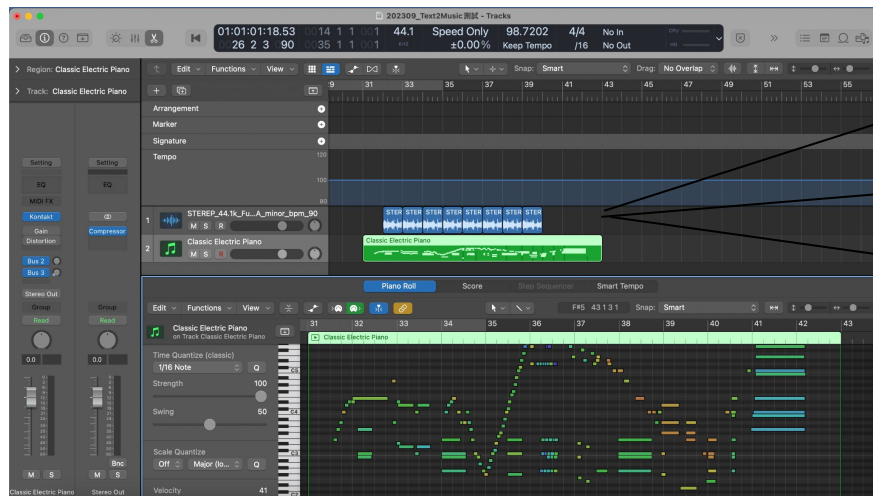


Virtual Pedalboard



Tape Machine

我們的試做



Guitar

Bass

Drum

Spatial Audio

空間與方位



Altiverb IR Browser



POST



domestic



car



aircraft-train
& boat



outdoor &
underground



next door
wall & floor



corridor
& stairs



public
spaces



school
& office

GEAR



plate, spring
digital

DESIGN



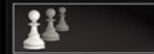
metal, glass,
toys & more

USER



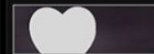
self made

THIRD PARTY



IR libraries

FAVORITES



your
favorites



opera
& theatre



stadium



mausoleum



club



misc

ALTIVERB

The original, professional convolution reverb

Another SPLICE?

Image-to-Reverb

輸入: Scene photo



AI Model

